Leukemia Survival Rate Prediction Using Machine Learning

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Abstract

This project investigates the use of machine learning methods to predict leukemia patient survival based on clinical and molecular features. A preprocessing and feature engineering pipeline was developed to prepare the data for modeling. Several algorithms were evaluated, including Random Survival Forest and XGBoost, with hyperparameters tuned via cross-validation. The Random Survival Forest achieved the best performance in terms of concordance index, highlighting the relevance of ensemble methods for survival prediction. Key predictors included medical center, white blood cell count, and cytogenetic risk.

1 Introduction

Leukemia is a hematological malignancy characterized by substantial heterogeneity in patient outcomes. Predicting survival is essential for guiding treatment decisions and improving patient management. In this study, we develop predictive models using clinical and molecular data, with a focus on building a robust and interpretable pipeline including preprocessing, feature engineering, model selection, and evaluation.

The full implementation of the preprocessing pipeline, feature engineering, and model training is available in the project repository. In particular, data preparation and feature construction are implemented in `DataManagement.py` and `ModelPipelines.py`, while model training, hyperparameter tuning, and evaluation are detailed in `GridSearch.py`. The model parameters for Random Survival Forest and XGBoost are detailed in `config.py`.

2 Dataset Description

The dataset combines clinical, hematological, cytogenetic, and genomic information for leukemia patients. Each patient is identified by a unique *ID* and described by both clinical and molecular features.

Clinical and Hematological Features

The dataset includes standard blood parameters reflecting disease severity and patient condition:

- *CENTER*: Treatment center, capturing potential differences in medical practices.
- *BM_BLAST*: Percentage of bone marrow blasts, a key marker of leukemia severity.
- *WBC*: White Blood Cell count (Giga/L), representing the total number of leukocytes in the blood.
- *ANC*: Absolute Neutrophil Count (Giga/L), an important indicator of immune system function.
- *MONOCYTES*: Monocyte count (Giga/L), reflecting a specific type of white blood cell involved in immune response.
- *HB*: Hemoglobin level (g/dL), measuring the oxygen-carrying capacity of the blood.
- *PLT*: Platelet count (Giga/L), essential for blood clotting and often affected in leukemia patients.

Cytogenetic Information

Cytogenetic abnormalities play a central role in leukemia prognosis and risk stratification.

- *CYTOGENETICS*: Detailed description of the patient's karyotype, following the ISCN standard. For instance, "46,XX" denotes a normal female karyotype, while abnormalities such as monosomy 7 (loss of chromosome 7) are associated with higher-risk disease.

Genomic and Mutation-Level Features

High-dimensional mutation data describe the molecular profile:

- *CHR*, *START*, *END*: Chromosomal location of the mutation on the human genome.
- *REF*, *ALT*: Reference nucleotide and alternate (mutated) nucleotide.
- *GENE*: Gene affected by the mutation.
- *PROTEIN_CHANGE*: Description of the amino acid modification induced by the mutation.
- *EFFECT*: Functional classification of the mutation (e.g., missense, nonsense), indicating its potential biological impact.
- *VAF*: Variant Allele Fraction, representing the proportion of sequencing reads carrying the mutation, and thus the prevalence of the mutation within the tumor cell population.
- *DEPTH*: Sequencing depth at the mutation site, corresponding to the total number of reads covering the genomic position.

Variable	dtype	count_nan	nan_%	count
CENTER	object	0	0.000000	11232
BM_BLAST	float64	420	3.739316	10812
WBC	float64	939	8.360043	10293
ANC	float64	685	6.098647	10547
MONOCYTES	float64	2049	18.242521	9183
HB	float64	382	3.400997	10850
PLT	float64	434	3.863960	10798
CYTOGENETICS	object	1408	12.535613	9824
CHR	object	411	3.659188	10821
REF	object	411	3.659188	10821
ALT	object	411	3.659188	10821
GENE	object	297	2.644231	10935
PROTEIN_CHANGE	object	309	2.751068	10923
EFFECT	object	297	2.644231	10935
VAF	float64	386	3.436610	10846
DEPTH	float64	411	3.659188	10821

Table 1: Dataset summary

Discussion of Data Structure

The dataset presents typical challenges of medical data. First, it mixes patient-level variables with mutation-level observations, resulting in multiple rows per patient. Second, it includes both numerical and high-cardinality categorical variables, especially in genomic features. Finally, missing values and heterogeneous scales require careful preprocessing.

These characteristics motivate the use of a feature engineering pipeline to aggregate mutation-level data at the patient level and prepare the dataset for modeling.

Exploratory Analysis

Correlation analysis highlights a strong positive relationship between *WBC*, *ANC*, and *MONOCYTES*, suggesting redundancy among immune-related variables. In contrast, most other features, including *HB*, *VAF*, and *DEPTH*, show weak correlations and may provide complementary information. A slight negative relationship is observed between *BM_BLAST* and *PLT*. Overall, the dataset contains one tightly connected cluster of immune cell variables alongside several relatively independent features.

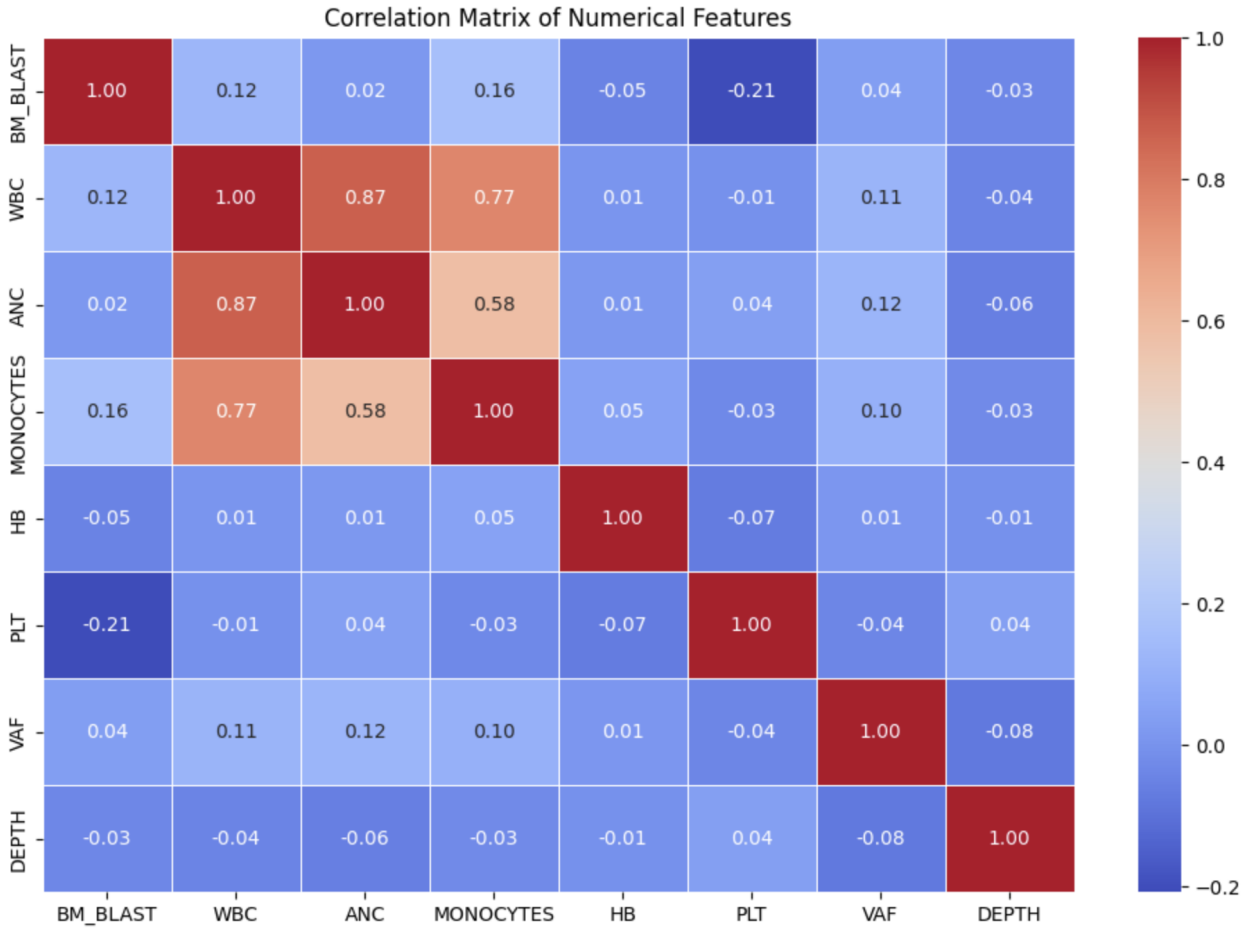

Figure 1: Correlation Matrix of Numerical Features

Survival outcomes (*OS_YEARS*) vary across centers, with similar median values but differing dispersion and outliers. Several centers (e.g., DUS, KI, and GESMD) display pronounced right-skewness with numerous high-value outliers, indicating the presence of long-term survivors, while others show more homogeneous outcomes. This variability likely reflects differences in patient populations, treatments, and data collection practices.

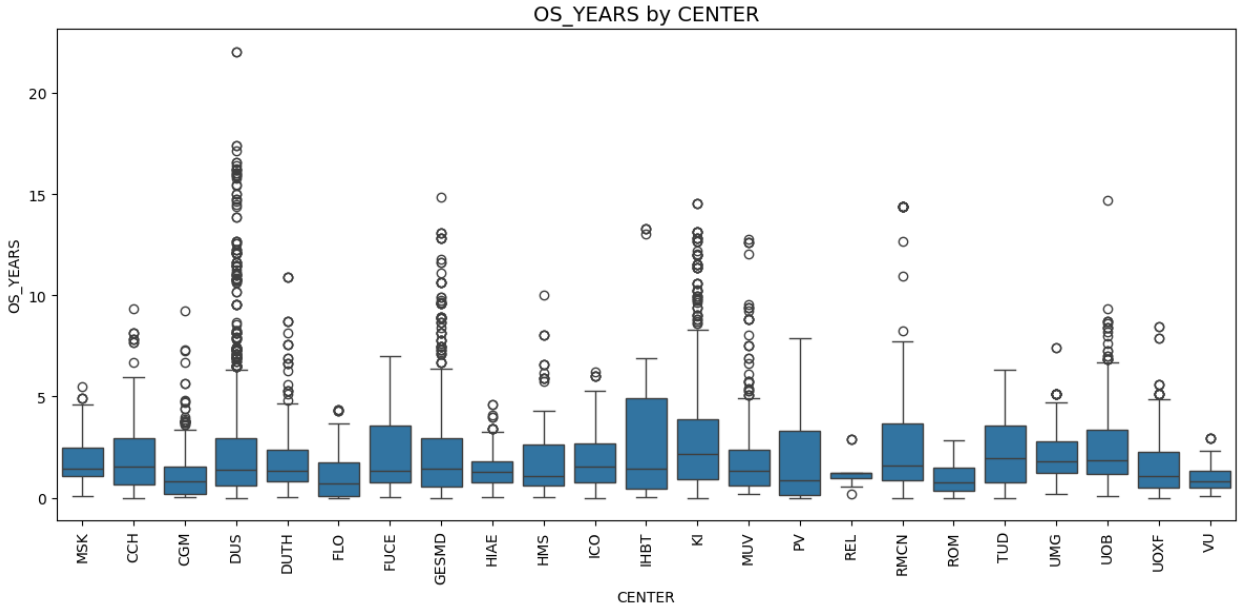

Figure 2: OS_YEARS by CENTER

3 Data Preprocessing and Feature Engineering

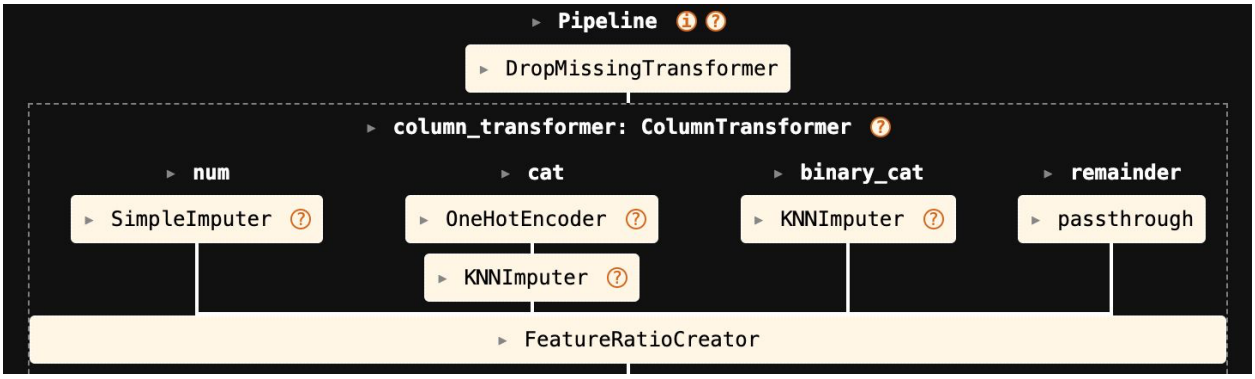

Figure 3: Preprocessing pipeline

The preprocessing pipeline was designed to handle heterogeneous medical data, including missing values, mixed feature types, and varying scales. All transformations were implemented within a unified pipeline to ensure reproducibility and avoid data leakage during cross-validation.

3.1 Handling Missing Values and Feature Selection

Features with more than 20% missing values were removed to limit unreliable imputations and reduce noise. The remaining preprocessing steps were implemented using a *ColumnTransformer*, which applies different transformations depending on the feature type.

3.2 Numerical Features

Numerical variables were processed using a *SimpleImputer* with a median strategy. Missing values were replaced by the median of each feature, computed on the training set. This choice is robust to outliers, which are common in medical data, and provides stable estimates of central tendency.

3.3 Categorical Features

Categorical variables were encoded using a *OneHotEncoder*, transforming them into a binary representation suitable for machine learning algorithms. Following encoding, a *KNNImputer* was applied to handle any remaining missing values. This transformation led to the creation of new features described in the below example, using 5 random patients.

ID	Initial CYTOGENETICS
P117425	46,xx,del(5)(q14q34),del(8)(q21q23)[6]/44-45,x...
P117426	46,xx
P117437	45,xx,der(4)del(4)(p13p15)del(4)(q23q26)der(5;...
P105454	44,xx,del(4)(p),del(5)(q13q33),der(7q),+8,add(...
P132707	complex

Table 2: Examples of raw cytogenetic annotations from the dataset. These karyotypes follow the ISCN nomenclature and may include complex chromosomal abnormalities.

3.4 Engineered Cytogenetic Features

CYTOGENETICS was transformed into binary indicators capturing clinically relevant abnormalities (e.g., *cyto_normal*, *cyto_complex*, *monosomy_7*, *trisomy_8*, *del_5q*). This transformation reduces the complexity of textual karyotypes while preserving key prognostic information.

- **cyto_normal**: Indicates the absence of detectable chromosomal abnormalities. Patients with normal karyotypes generally have more favorable outcomes.
- **cyto_complex**: Represents genomic instability, where multiple chromosomal abnormalities are present. Such patients are typically treated more aggressively due to higher-risk disease.

- **monosomy_7**: Denotes the loss of a chromosome in the 7th pair, affecting genes involved in hematopoiesis. This abnormality is associated with poor prognosis.
- **trisomy_8**: Marks the presence of an extra copy of chromosome 8, which can influence disease progression and risk stratification.
- **del_5q**: Indicates the deletion of the long arm of chromosome 5. Patients with this alteration often respond well to specific therapies such as lenalidomide.
- **t_3_3**: Represents a translocation between regions of chromosome 3, commonly associated with aggressive disease and poor prognosis.
- **cyto_mosaic**: Captures the presence of multiple cell populations within a patient, reflecting clonal heterogeneity. This feature helps understand tumor evolution and potential treatment resistance.
- **number of abnormalities**: A numerical feature approximating the overall cytogenetic complexity. Higher counts generally correlate with worse prognosis.

ID	cyto_normal	cyto_complex	monosomy_7	trisomy_8	del_5q	t_3_3	cyto_mosaic
P117425	0	0	1	0	1	0	1
P117426	1	0	0	0	0	0	0
P117437	0	0	0	0	0	0	1
P105454	0	0	0	1	1	0	0
P132707	0	1	0	0	0	0	0

Table 3: Example of engineered cytogenetic features derived from raw ISCN annotations. Each variable represents the presence (1) or absence (0) of clinically relevant chromosomal abnormalities.

3.5 Genomic Feature Aggregation

Mutation-level data were aggregated at the patient level by constructing a matrix of Variant Allele Frequencies (VAF) per gene. Each feature represents the presence and intensity of mutations in a given gene.

ID	GENE	VAF
P117425	DNMT3A	0.0330
P117425	TP53	0.2130
P117425	DNMT3A	0.3026
P117426	DDX23	0.0270
P117426	SF3B1	0.3010
P117426	MLL	NaN
P117437	TET2	0.5860
P117437	TP53	0.4000
P105454	TP53	0.8490
P105454	DNMT3A	0.4530
P132707	TP53	0.6900

Table 4: Examples of raw genomic features, including gene names (*GENE*) and variant allele fractions (*VAF*) for each patient. Multiple entries per patient correspond to distinct mutations observed in their genomic profile.

ID	DNMT3A	TP53	DDX23	SF3B1	MLL	TET2	TP53 (dup)
P117425	0.3356	0.213	0.000	0.000	0.000	0.000	0.213
P117426	0.0000	0.000	0.027	0.301	0.000	0.000	0.000
P117437	0.0000	0.400	0.000	0.000	0.000	0.586	0.400
P105454	0.4530	0.849	0.000	0.000	0.000	0.000	0.849
P132707	0.0000	0.690	0.000	0.000	0.000	0.000	0.690

Table 5: Example of engineered genomic features representing Variant Allele Frequency (VAF) per gene. Each value reflects the proportion of sequencing reads supporting a mutation in the corresponding gene for a given patient.

3.6 Additional Feature Engineering

Additional features were created using ratios of biological variables (e.g., *ANC/WBC*, *PLT/WBC*). Such transformations aim to capture non-linear relationships and relative effects that are not directly observable from raw features.

The final dataset contains more than 150 features.

3.7 Discussion of the Pipeline

The use of a unified pipeline ensures that all transformations are fitted only on the training data and consistently applied to validation and test sets, preventing data leakage.
Overall, this approach enables the integration of heterogeneous data sources while producing a structured and informative feature space suitable for machine learning models.

4 Machine Learning Models

We focus on tree-based ensemble methods, which are well-suited for structured and heterogeneous medical data. In particular, we consider Random Survival Forest and XGBoost, both capable of capturing non-linear relationships and feature interactions.

4.1 Random Forest

Random Survival Forest (RSF) extends Random Forest to survival analysis by handling censored data and time-to-event outcomes. Each tree is trained on a bootstrap sample, and splits are chosen to maximize survival differences between nodes.

Predictions are obtained by aggregating survival estimates across trees, typically in the form of a risk score or survival function.

RSF is particularly appropriate in our setting as it handles censored data naturally, captures non-linear effects and interactions and is robust to noise and high-dimensional features.
Key hyperparameters include the number of trees, maximum depth, and minimum node size, tuned via cross-validation.

4.2 XGBoost

XGBoost is a gradient boosting method that builds trees sequentially, each correcting the errors of the previous ensemble. It optimizes a regularized objective function combining a loss term and a complexity penalty.

Its main strengths include: strong predictive performance, built-in regularization and ability to handle missing values.

Hyperparameters such as learning rate, maximum tree depth, and number of boosting rounds were tuned using cross-validation.

4.3 Model Evaluation

Both models were evaluated within the same preprocessing pipeline to ensure consistency. Hyperparameters were optimized using cross-validation.

Performance was primarily assessed using the concordance index (C-index), which measures the ability of the model to correctly rank survival times. This metric is standard in survival analysis and accounts for censored observations.

This framework allows a comparison between bagging-based (RSF) and boosting-based (XGBoost) approaches for survival prediction.

The implementation of both Random Forest and XGBoost models, along with the associated hyperparameter tuning procedures, is available in the repository (see `config.py`).

5 Model Training and Evaluation

The dataset was split into training and testing sets using an 80/20 ratio. All preprocessing steps were fitted exclusively on the training data and then applied to the test set to prevent data leakage.

Hyperparameters were optimized using grid search combined with 5-fold cross-validation, providing a robust estimate of out-of-sample performance.

5.1 Evaluation Metric: Concordance Index (C-index)

Model performance was primarily assessed using the concordance index (C-index), which measures the ability to correctly rank patients by risk in the presence of censored data. We do not use F1-score as it is not appropriate in such context because it does not account for the time-to-event dimension nor for censored observations.

Definition
The C-index measures the proportion of all comparable pairs of patients for which the predicted risks are correctly ordered with respect to their observed survival times. Formally, it is defined as:

$$C = \frac{\text{Number of concordant pairs}}{\text{Number of comparable pairs}} \tag{1}$$

Interpretation
Consider two patients i and j:

- If patient i experiences the event (e.g., death) before patient j, then a well-performing model should assign a higher predicted risk to patient i, i.e., $\hat{r}_i > \hat{r}_j$.
- A pair is said to be *concordant* if the predicted risks are correctly ordered.
- A pair is *discordant* if the ordering is incorrect.
- A pair is considered *tied* if both patients have the same predicted risk.

A value of 0.5 corresponds to random predictions, while values above 0.7 indicate good discriminative ability.

Results
The Random Survival Forest achieved a C-index of 73.24%, while XGBoost slightly outperformed it with 73.68%.

Both models therefore exhibit similar and good discriminative performance, correctly ranking patient risk in approximately 73% of comparable cases. The performance gap is small, suggesting that both approaches capture similar patterns in the data.

5.2 Learning Curve

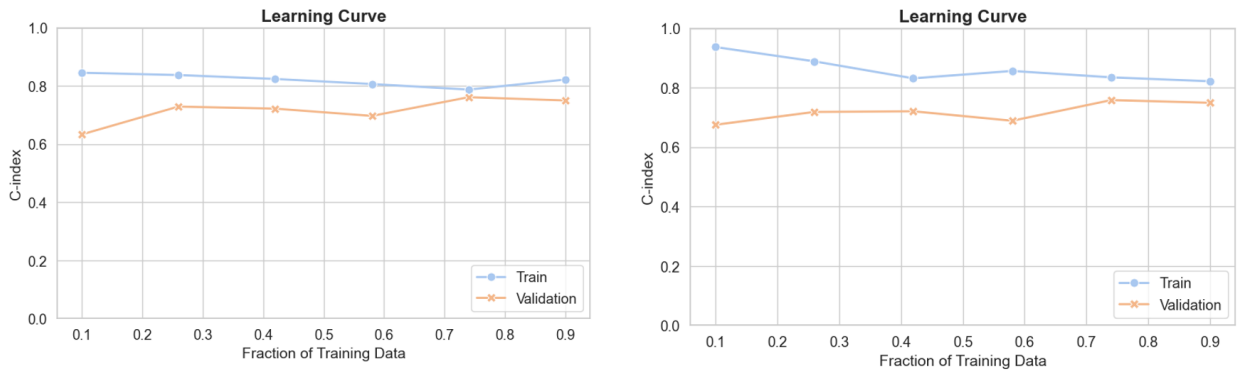

Figure 4: Learning Curve: RSF vs. XGBoost

XGBoost shows higher training performance for small sample sizes, with a C-index close to 0.95 at low fractions of the training data, but with a larger gap between training and validation scores. This suggests stronger overfitting. In contrast, RSF exhibits a smaller and more stable gap between training and validation curves, suggesting better robustness.

As the training size increases, both models converge to similar validation performance (around 0.75), consistent with their close C-index values. Notably, the gap between training and validation curves decreases more significantly for XGBoost, indicating that additional data helps mitigate its initial overfitting behavior.

5.3 Risk Score Distribution

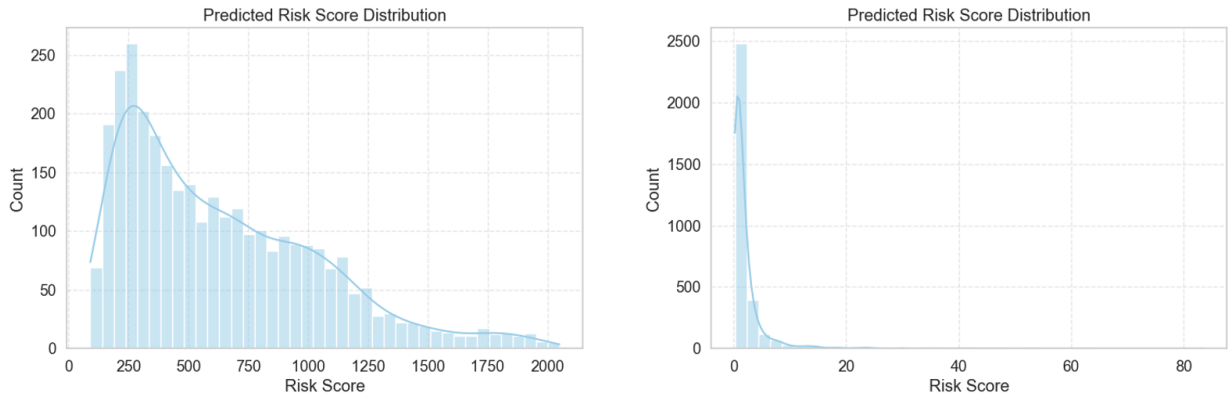

Figure 5: Predicted Risk Score Distribution: RSF vs. XGBoost

RSF produces a wider and more dispersed distribution of risk scores, indicating finer differentiation between patients. The distribution is moderately right-skewed, indicating that while most individuals fall into lower-to-mid risk categories, there is a substantial spread toward higher-risk values. In contrast, XGBoost generates highly concentrated and sharply right-skewed predictions, with most scores clustered in a narrow range.

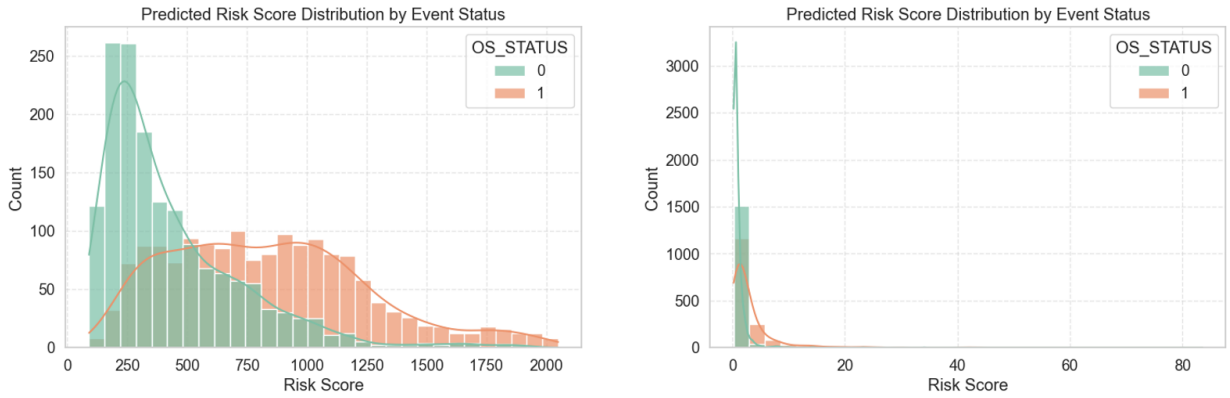

Figure 6: Predicted Risk Score Distribution by Event Status: RSF vs. XGBoost

RSF shows clearer separation between event and non-event groups, suggesting better discrimination. XGBoost exhibits more overlap, which may limit its ability to distinguish between risk levels.

6 Discussion

The results highlight the effectiveness of ensemble methods for predicting leukemia survival from heterogeneous clinical and genomic data. Random Survival Forest (RSF) and XGBoost achieve similar performance in terms of C-index, indicating comparable ability to capture underlying patterns.

However, their behaviors differ. XGBoost shows higher training performance for small sample sizes but exhibits a larger gap between training and validation scores, suggesting a greater tendency to overfit. In contrast, RSF demonstrates more stable generalization, which is advantageous in limited-data settings typical of medical applications.

The analysis of predicted risk scores provides additional insight. RSF produces a wider and more dispersed range of values, allowing finer differentiation between patients. XGBoost, by contrast, generates more concentrated predictions, which may reduce its ability to distinguish between close risk levels.

From a feature perspective, combining clinical, cytogenetic, and genomic data proves essential. In particular, transforming cytogenetic annotations and aggregating mutation-level data into structured features significantly improves model usability and performance.

Several limitations should be noted. First, heterogeneity across clinical centers may introduce biases related to treatment practices or data collection. Second, aggregating mutation-level data at the patient level may lead to a loss of information. Finally, the relatively small sample size may limit the performance of more flexible models such as XGBoost.

Overall, both models perform well, with XGBoost achieving slightly higher predictive accuracy and RSF providing more stable and interpretable risk estimates. This highlights a trade-off between predictive performance and robustness in survival modeling.

7 Conclusion

This study demonstrates the effectiveness of machine learning methods for predicting leukemia survival using combined clinical and genomic data. A structured preprocessing and feature engineering pipeline enabled the transformation of heterogeneous and partially incomplete data into a format suitable for modeling.

Both Random Survival Forest and XGBoost achieved similar predictive performance, highlighting their ability to capture complex relationships in the data. While XGBoost obtained slightly higher C-index values, RSF provided more stable generalization and more interpretable risk estimates, illustrating a trade-off between flexibility and robustness.

The results also underline the importance of feature engineering, particularly the transformation of cytogenetic and genomic data into structured representations, which proved essential for effective learning.

Future work could incorporate additional data sources, such as treatment information or longitudinal follow-up, and validate the models on external cohorts to improve robustness and clinical applicability.

Overall, this project shows how combining domain knowledge, careful preprocessing, and appropriate machine learning methods can support survival prediction and contribute to more personalized approaches in leukemia care.